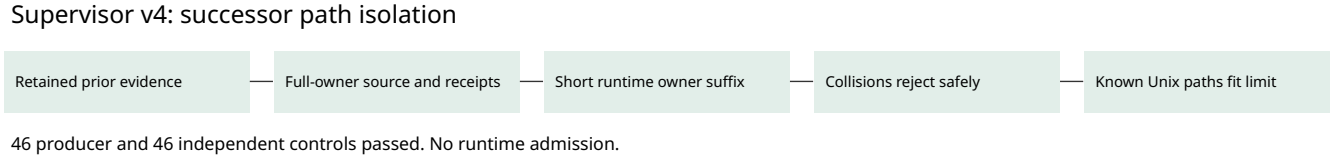
Supervisor v4: retained-disk successor paths

Qualification source snapshots and receipts now use the full owner UUID. A new owner gets fresh directories while the failed predecessor evidence remains intact. Existing source directories fail at mkdir, and existing receipt paths, including dangling symlinks, are rejected before copying. The callback receives the exact pinned source and binding files plus the new receipt path.



The path audit also found that e-q and s-q collided on a second qualification over the retained disk. Qualification exec, shim and Docker socket names now include the first eight hexadecimal owner characters. This suffix is not assumed unique: two distinct UUIDs sharing it are rejected by existing-path checks, preserving the old bytes. A retained Docker socket path or dangling symlink is rejected before daemon startup. Assembly names remain unchanged.

Path bounds. Known qualification Docker, containerd, debug, ttrpc and shim pathname lengths are 71, 93, 99, 99 and 82 bytes. Each plus its terminating NUL fits the Linux 108-byte sun_path. Full UUID source and receipt directories contain ordinary files, not these Unix sockets. Known derivations are checked; this is not an exhaustive runtime socket census.

Other fixed paths. The Docker data store is intentionally retained under the existing disk custody and lock checks. Context, base-input archive and runtime.iid are assembly-only. Daemon config, pid, logs, client config and command output use full owner identifiers. The outer root and service cgroup also use full owners. The /run/containerd mask remains private to each service namespace and binds its current shim directory. No deletion or reuse of failed evidence was introduced.

Validation. All 46 mocked/local-temporary controls passed: 36 retained controls and 10 new successor checks. The new checks cover separate owners, same-owner collision, a shared short prefix, retained sockets and symlinks, pinned source substitution, exact callback paths, prior-evidence preservation and known socket lengths. The single producer wrapper was bounded by 120 seconds including kill reserve and 2 GiB address space. Independent source review accepted the exact source and replayed all 46 controls. A short-prefix collision can leave the new owner's client-config directory, while preserving the previous owner's evidence. No service, mount, daemon, gate or qualification was executed.